Incident Report

Lateral Movement & Service Execution Simulation

Framework Mapping: MITRE ATT&CK

Techniques: TA0008, TA0002

Date: October 2026

Analyst: Security Team

1. Executive Summary

This document details a simulated internal network compromise designed to test detection capabilities against lateral movement and remote service execution. The objective was to emulate an adversary leveraging compromised credentials to pivot across a network using Server Message Block (SMB) and Windows administrative shares.

The attack was executed from a Kali Linux attack machine targeting a Windows host. Following the simulated intrusion, forensic evidence was collected locally from the victim machine and ingested into Splunk Enterprise to hunt for distinct Indicators of Compromise (IoCs). The analysis successfully correlated network logons (Type 3) with background service logons (Type 5) and malicious service creation, providing a high-fidelity detection pattern.

2. Attack Execution

The adversary utilized the **Impacket** suite, specifically the `psexec.py` module, to authenticate to the target machine (172.16.234.131) via SMB. By passing valid credentials, the tool uploaded a randomized executable payload to the `ADMIN$` share, registered it as a Windows service, and executed it to spawn an interactive system-level shell.

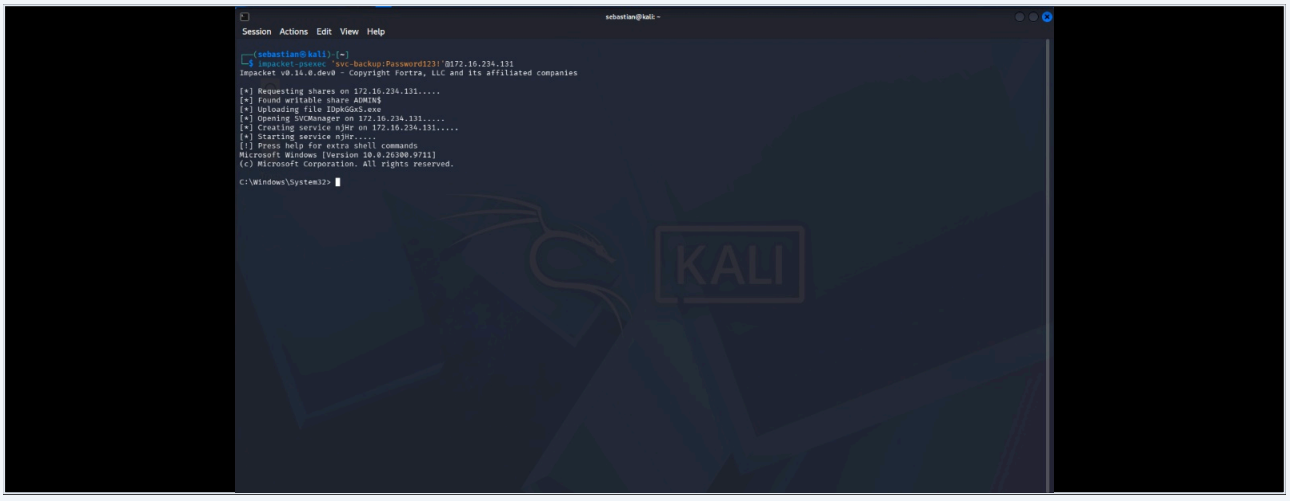


Figure 1: Successful execution of the psexec payload from the Kali Linux terminal, resulting in a remote Windows command prompt.

3. Evidence Collection (Local Acquisition)

To preserve forensic integrity, log acquisition was performed directly on the target host immediately following the simulated attack. The Windows Event Viewer was utilized to isolate the relevant audit trails necessary for detecting the adversary's actions.

Both the Windows **Security** logs (recording authentication and access control) and the **System** logs (recording service control manager events) were exported as flat `.csv` files for seamless ingestion into the Splunk SIEM environment.

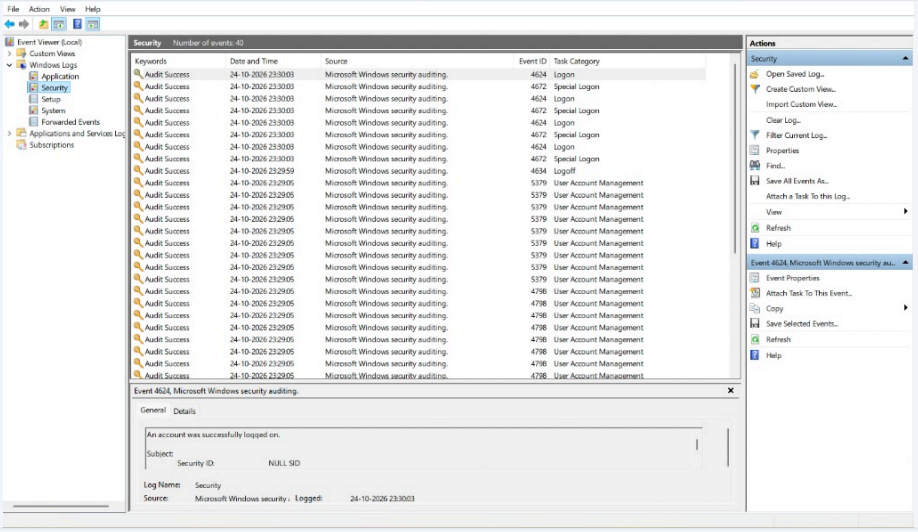


Figure 2: Exporting the Windows Security log events (including Event ID 4624) from the compromised host.

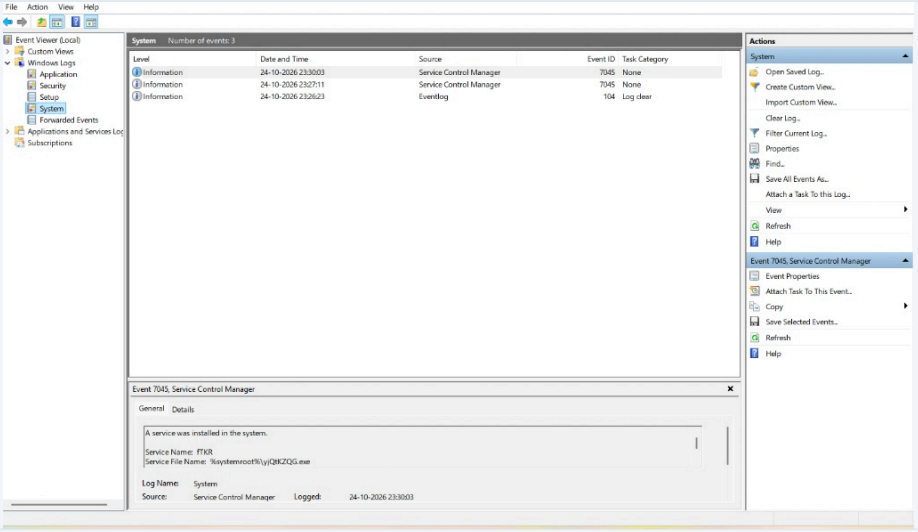


Figure 3: Exporting the Windows System log events (including Event ID 7045) generated by the Service Control Manager.

4. SIEM Forensic Analysis (Splunk)

The exported CSV data was ingested into Splunk Enterprise to conduct a centralized threat hunt. The primary objective was to trace the attacker's footprint from initial network authentication through to the execution of the payload.

4.1 Network Authentication Verification

The first indicator of the psexec attack is the remote SMB connection. The Security logs were queried for successful logons (Event ID 4624) specifically occurring over the network.

```
source="security.csv" "Event ID"=4624 | regex _raw="Logon Type:\s+3"
```

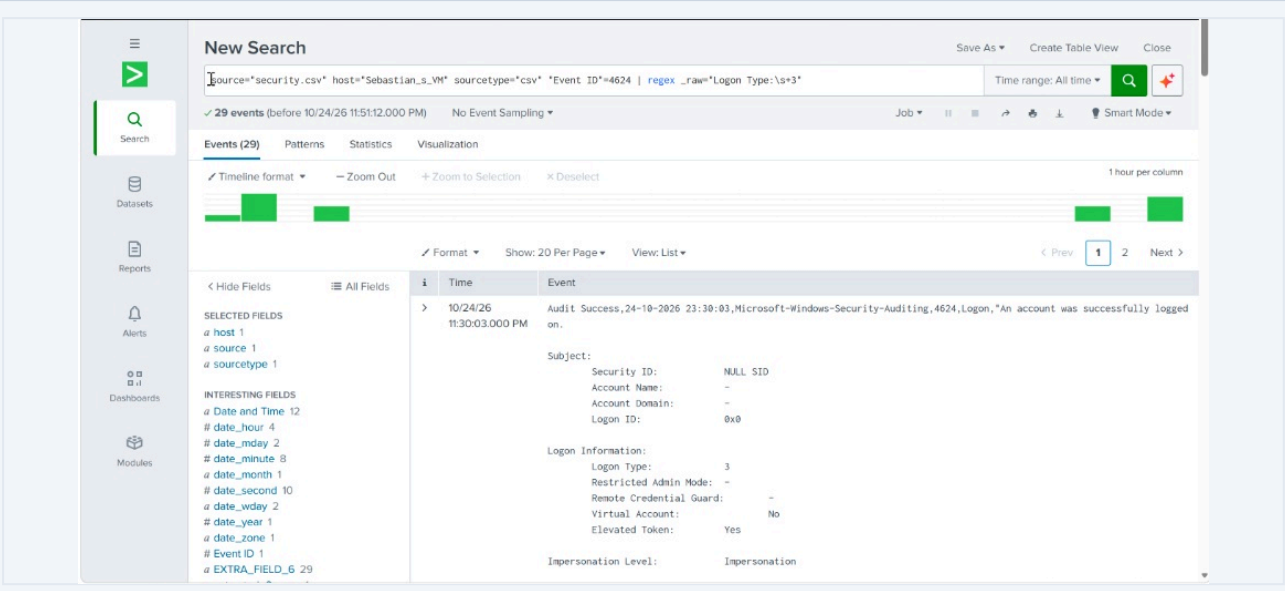


Figure 4: Splunk isolating a Logon Type 3 event, confirming external network authentication to the target.

4.2 Background Service Authentication

To execute the payload silently, the attacker's tool leverages the Windows Service Control Manager. This generates a secondary authentication event specifically for the service account.

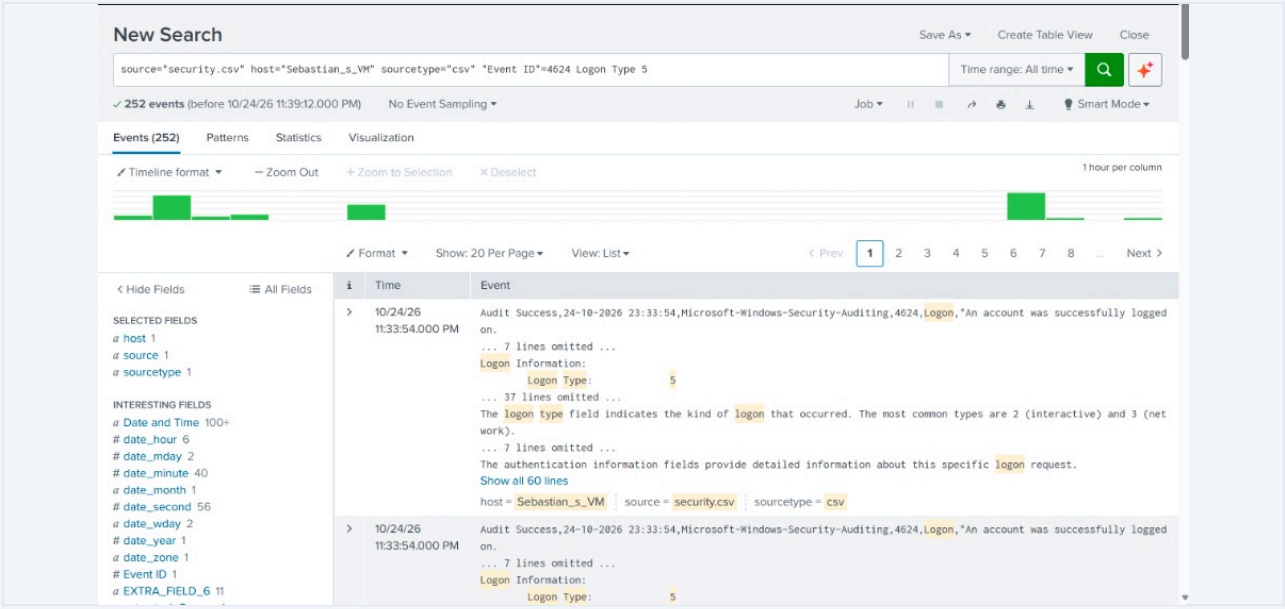


Figure 5: Splunk isolating a Logon Type 5 event, proving a background service was authorized to run immediately following the network logon.

Analysis:

The observed Logon Type 5 event indicates a Windows service logon associated with the service execution phase. When correlated with the preceding Logon Type 3 network authentication and the subsequent Event ID 7045 service-creation event, it strengthens the evidence of possible PsExec-based lateral movement. However, this event alone is not conclusive evidence of malicious activity.

4.3 Malicious Service Creation (Payload Execution)

The most definitive artifact of this lateral movement technique is the registration of the randomized executable as a new Windows service. The System logs were queried for Event ID 7045 (A service was installed in the system) to identify the exact payload.

```
source="system.csv" "Event ID"=7045
```

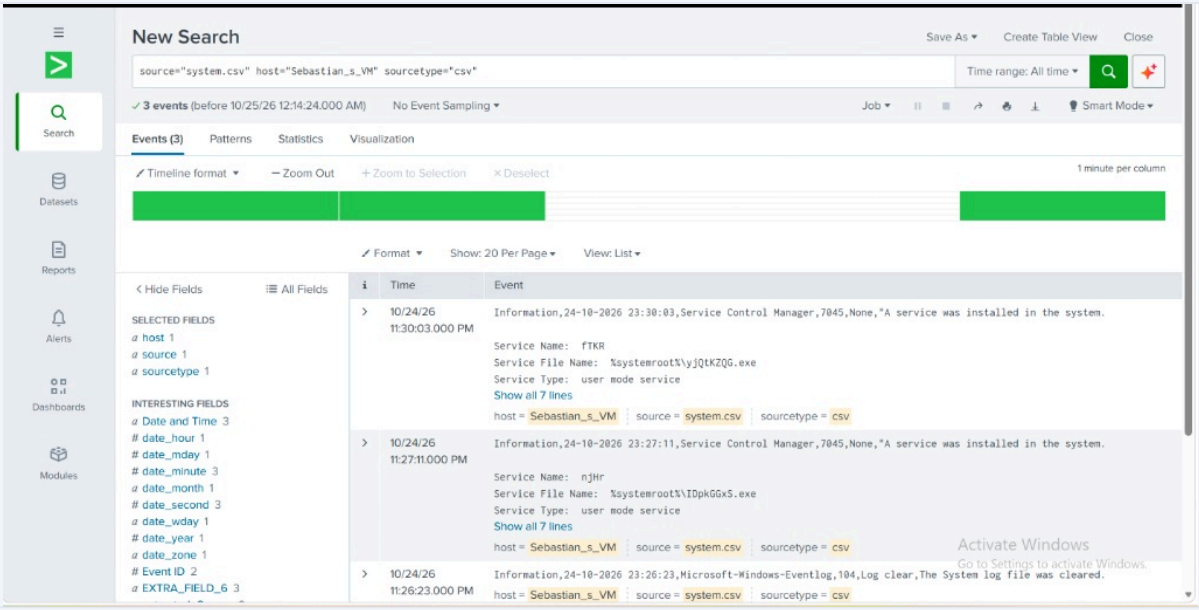


Figure 6: Splunk uncovering the exact malicious service names (fTKR, njHr) and their corresponding executable paths dropped by the psexec utility.

Conclusion:

The forensic timeline confirms a successful lateral movement compromise. The sequence of a Type 3 Network Logon, immediately followed by a Type 5 Service Logon and an Event ID 7045 Service Creation event mapping to a randomized executable in the `%systemroot%` directory, is a highly reliable signature of Impacket/psexec activity. It is recommended that automated alerting rules be configured in the SIEM to trigger whenever these three events occur in rapid succession on a single host.